

Modèle Conceptuel de Données (MCD) – Version 1.0 du projet MyDigitrans.

Cette version est optimisée pour :

- Multi-agence
- Multi-devise
- Paiement mixte
- Gestion caisse
- Réservation / Paiement partiel / Billet final
- Module colis
- Module commission
- Historique simple sécurisé

1. **ORGANISATION**

1. **Agency**

- Name
- resgistrationNumber(*nullable*)
- taxNumber(*nullable*)
- Phone
- Email
- Address
- isActive
- Country
- City
- createdAt
- UpdatedAt(*nullable*)
- Slug
- Code
- Logo (*nullable*)
- DefaultCurrency(*nullable*)

2. **Branch**

- Name
- Agency
- City
- Country
- Status (*opened, closed, suspended*)
- Code
- Slug
- createdAt
- updatedAt(*nullable*)
- isActive

3. **User**

- firstname
- lastname
- phone
- email
- password
- userType(*platform, agency, partner*)
- sponsor (user)
- isActive
- isOnline
- code
- lastLoginAt
- agency (*nullable*)
- brach(*nullable*)
- createdAt

4. **Role (system)**

- name
- code (*ROLE_SUPER_ADMIN, ROLE_CASHIER,...*)
- scope (*platform, agency*)
- description(*nullable*)
- isActive

5. **UserRole**

- user
- role
- agency(*nullable*)
- branch(*nullable*)
- assignedAt
- AssignedBy
- RevokedAt(*nullable*)
- revokedBy(*nullable*)

6. Permission (system)

- name (*lecture ticket*)
- code (ticket.view)
- description(*nullable, voir la liste des tickets vendu par exemple*)
- module(ticket,payment,agency,branch,trip etc.)
- createdAt
- createdBy
- scope(platform,agency)

7. rolePermission

- role
- permission
- createdAt
- branch(nullable)
- agency(nullable)
- assignedAt
- AssignedBy
- RevokedAt(nullable)
- revokedBy(nullable)

Role	Permission
Cashier	Ticket.create
Cashier	Ticket.view
Manager	Ticket.refubd

2. TRANSPORT

8. busLayout (System)

- name (*modèle 2-2 ; ; modèle 3-1 ; etc*)
- agency(*nullable*)
- rows
- columns
- aisles(*Json position de couloirs ex : [" 2"] un couloir apres la deuxieme colonne*)
- specialPositions (*JSON*)
- hasBackExtraSeat(*bool*)
- isValidated
- createdAt

9. Bus

- agency
- busLayout
- code (nullable)
- name(nullable)
- plateNumber(unique)
- brand (toyota, mercedes etc)

- model (Tx, Maybach, etc)
- status(*active,maintenance,inactive*)
- inactiveReason
- mileage
- year
- createdAt
- createdBy

9 BusMaintenance

- bus
- issuedAt
- reportedAt
- reportedBy
- resolvedAt
- updatedBy
- solution
- severity(*critical,medium,low*)
- mileage(nullable)
- createdAt
- isCancelled
- statut(*broken,resolved*)

10. BusDocuments

- bus
- type (Assurance, contrôle technique,...)
- referenceNumber (nullable)
- issuedAt
- expiredAt
- updatedAt
- updatedBy
- status(*valid,expired*)
- url(nullable)
- code (unique)
- createdAt
- createdBy

10. Seat (automatique creation)

- bus
- seatLabel
- type(*driver, normal, vip, extra,wc*)
- row
- column
- createdAt

11. Route

- agency
- name

- code
- slug
- distance (km)
- duration
- createdAt
- createdBy
- isAvailable
- lastUpdatedAt
- Reservation_expiration_delay(
*nullable et en minute ex : 120
pour deux heures*)
- Price_per_kg(*ex 500 pour dire
500CDF par kilo si la devise
c'est CDF=>10kg = 5000CDF*)
- Currency_price_per_kg

12. routeStop

- route
- stopOrder,
- city,
- country,
- distanceFromDeparture,
- durationFromDeparture,
- isActive,
- createdAt,
- createdBy

13. Trip

- route
- bus
- departureDate
- departureTime
- arrivalDate
- arrivalTime
- driver(user)
- coDriver(nullable user)
- status(*scheduled,boarding,departed,arrived,cancelled*)
- code
- createdAt
- createdBy

14. TripSeat (*automatique after trip creation*)

- unique(trip,seat to avoid
duplication)
- trip
- seat
- seatStatus(*held, reserved,
blocked, available*)
- reservation(nullable)
- heldUntil(nullable)

3 CLIENT



Reconvertie en user (donc customer est
supprimé car les clients auront des comptes
plutard pour suivre leurs billets etc)



4 TARIFICATION

15. Currency

- code (USD, CDF) (unique)
- name (*dollar, Franc Congolais*)
- agency(nullable)
- createdAt
- createdBy

16. ExchangeRate(agency admin)

- baseCurrencyId
- targetCurrencyId
- rate
- createdAt
- agency(nullable)
- effectiveFrom

17. Fare

- fareType (*promo, standard, VIP*)
- targetType (Ex: "route", "subscription_plan",
"feature")
- targetId
- departureStop
- arrivalStop
- isActive
- price_per_kg
- agency

- amount
- currency
- validFrom
- validUntil
- createdAt
- createdBy

5. **VENTE BILLET**

18. Reservation

- reference
- trip
- departureStop(routeStop)
- arrivalStop(routeStop)
- agency
- user
- status(*PENDING, CONFIRMED, CANCELLED, EXPIRED*)
- totalAmount
- currency
- createdAt
- createdBy
- expiresAt
- cancellationReason
- canceledBy
- **PENDING** : Reserved but not paid
- **TOTALAMOUNT** : Je reserve par exemple 3 siege de 15\$ le total sera 45\$
- **ExpiresAt** : interaction avec tripseat, si la reservation expire, le seat devient available

19. Ticket

- Agency
- ticketNumber
- Reservation
- tripSeat
- travelerName
- travelerPhone
- price
- currency
- status (*active, cancelled, refunded, used*)
- createdAt
- createdBy

20. AgencyPayment

- type (*reservation, Shipment*)
- reference
- amount
- currency
- exchangeRateUsed
- reservation(nullable)
- shipment(nullable)
- paymentMethod (*cash, mobile_money, card, bank_transfert*)
- status(*pending, failed, refunded, confirmed*)
- cashSession (FK CashSession, nullable pour les paiements en ligne)
- mobileMoneyReference(nullable)
- createdBy (nullable)
- confirmedBy (nullable)
- confirmedAt
- createdAt
- agency
- branch

21. Refund

- payment
- agency
- amount
- currency
- reason
- status(*pending, success, failed*)
- createdBy
- createdAt

22. Discount

- referencedType (*reservation, shipment*)
- referencedId
- amount
- currency
- createdAt
- createdBy

23. TicketPrintLog

- ticketId
- printedBy
- printedAt
- reason (*original, duplicate*)
- printCount

24. TicketScanLog

- ticketId
- scannedBy
- scannedAt
- result (valid, already_used, invalid)

6. MODULE COLIS

25. Shipment

- reference
- tripId
- senderId (userId)
- receiverId (userId)
- description
- nature
- departureAgencyBranch
- arrivalAgencyBranch
- weight
- billing_type (*weight, unit*)
- price
- currency
- agency
- branch
- status (Pending, in_transit, ready_for_pickup, delivered, canceled)
- departureStop (RouteStop)
- arrivalStop (RouteStop)
- createdBy
- createdAt

7. CAISSE

26. CashRegister

- agency
- branch
- name (Guichet 1, etc)
- code
- currency
- status (*active, inactive*)
- hardware (nullable, les matériels informatique assignés)
- createdAt
- createdBy

27. CashSession

- cashRegister
- openedBy
- openedAt
- closedBy
- closedAt
- status(open, closed, pending)

28. CashSessionBalance

- cashSession
- currency
- openingBalanceExpected
- openingBalanceDeclared
- closingBalanceExpected
- closingBalanceDeclared
- openBalanceDifference
- closingBalanceDifference
- openingObservation (pour expliquer le manque par exemple)
- closingObservation
- managerConclusion (Pour la cloture avec ecart)
- openingNote (consigne à l'ouverture)

28. Expense

- type (*subscription, other*)
- name (*abonnement mars 2026, transport agents, etc*)
- amount

- currency
- cashRegister (*nullable*)
- agency
- branch
- description
- status (*pending, paid, failed*)
- validFrom
- validUntil
- createdAt
- createdBy
- confirmedBy

29. FinancialOperation

- cashSessionId
- type (*in, out*)
- referenceType(*agencyPayment, refund, expense*)
- referenceId
- amount
- currencyId
- createdAt
- createdBy
- description

9. SOUSCRIPTION

30. Feature

- code
(*online_booking, advanced_reports, shipment_modu
le etc*)
- name (*Reservation en ligne, Gestion colis etc*)
- description
- status(*active, inactive*)
- createdAt
- createdBy

31. subscriptionPlan

- name (*basic, pro, trial etc*)
- price
- currency
- duration_days
- max_branches
- max_users
- max_buses

- status (*active, inactive*)
- createdAt
- createdBy

32. PlanFeature

- plan
- feature
- createdAt
- createdBy

33. subscription

- agency
- plan
- startAt
- endAt
- status (*active, expired, canceled, suspended*)
- createdAt
- createdBy

34. Invoice

- agency
- type (*subscription, platformFee, mixed, donc abonnement, commission sur les ventes et les deux*)
- invoiceNumber
- amount
- currency
- status
(*draft, issued, paid, cancelled, overdue*)
- issuedAt (*facture envoyée le*)
- dueDate(*date limite sinon le status change par ex : expired*)
- paidAt
- createdAt
- createdBy
- subtotal
- discountTotal
- total
- balanceDue (*reste à payer pour le cas de paiement partiel*)
- periodStart
- periodEnd

35. platformFeeRule

- sourceType(reservation,shipment)
- feeType (fixed,percent)
- value
- currency (nullable pour le pourcentage)
- agency (nullable, si fixé, cette agence suivra cette rule)
- createdAt
- createdBy

36. platformFee (créer à la confirmation du paiement de la reservation ou colis)

- agency
- rule
- sourceType(reservation,shipment)
- sourceId
- amount (si fixe rule.value, si pourcentage $\text{ticket.price} * \text{rule.value} / 100$)
- currency
- status (pending,invoiced,paid, partially_paid)
- invoice(nullable)
- createdAt

37. platformRefund

- invoice
- agency
- amount
- currency
- reason
- status(pending,success,failed)
- createdBy
- createdAt

38. PlatformDiscount (definit les promotions)

- type (fixed, percent)
- name (promo, bonus)
- value
- currency(nullable)
- validForm

- validUntil
- maxUses (limite d'utilisation)
- usedCount (nombre de fois utilisé)
- status (active,inactive,expired)
- createdAt
- createdBy

39. InvoicePlatformDiscount

- invoice
- amount
- currency
- platformdiscount (nullable)

40. subscriptionPayment

- subscription
- invoice
- amount
- currency
- method (manual, mobile money etc)
- paidAt
- status (pending,success,failed,refunded)
- refrence
- platformCashSession(platformCashSession)
- confirmedBy (nullable)
- createdAt
- createdBy
- confirmedAt (nullable)

9. **PLATFORME CAISSE**

41. PlatformBranch

- name
- city
- adress
- status
- createdAt
- createdBy

42. platformCashRegister

- platformbranch
- name
- code
- currency
- status

- createdAt
- createdBy

43. platformCashSession

- platformCashRegister
- platformbranch
- openedBy
- openedAt
- closedBy
- closedAt
- openingBalance
- closedBalance
- countedBalance
- difference
- status (*closed, opened*)

44. PlatformExpense

- name(*hebergement, loyer, transport agents, etc*)
- amount
- currency
- platformbranch
- description
- status (*pending, paid, failed*)
- validFrom
- validUntil
- createdAt
- createdBy
- confirmedBy

45. platformFinancialOperation

- platformCashSession
- operationType (*commission, subscription_payment, refund, expense, adjustment*)
- type (*in, out*)
- referenceType(*invoice, refund, commissionPayment, platformExpense*)
- referenceId
- platformCashSession(platformCashSession)
- amount
- currencyId
- createdAt
- createdBy
- description

10. COMMISSION

46. commissionRule

- name
- sourceType (*Ticket, Shipment, subscription*)
- level
- maxLevel
- percentage
- isActive
- createdAt
- createdBy

47. commission

- user
- sourceType
- sourceId
- level
- percentage
- amount
- currency
- createdAt
- status (*pending, approved, paid, cancelled*)
- approvedBy (*nullable*)
- approvedAt (*nullable*)

48. commissionPayment

- user
- amount
- currency
- method
- reference
- status (*pending, success, failed*)
- paidAt
- createdAt
- createdBy
- paidBy (*agency, platform*)
- paidById (*user*)

49. commissionPaymentItem

- commission (*unique*)
- commissionPayment
- amount

- createdAt
- user_agent(Mozilla 5.0 ; android chrome etc)

11 SÉCURITÉ & HISTORIQUE

50. AuditLog

- user
- action (*create, update, cancel, refund, login, delete, confirm_payment, cancel_booking*)
- entityName (resevation etc)
- entityId
- oldValues(*JSON['status': 'pending']*)
- newValue (*JSON['status': 'success']*)
- createdAt
- ipAddress
- device_type(mobile, POS, Desktop)

🔗 LOGIQUE DOCUMENTAIRE

Un seul Ticket en base.

Selon paymentStatus :

- unpaid → Bon de réservation
- partial → Reçu de paiement
- paid → Billet officiel